

Heart Disease Analysis

1st PDF - Preprocessing Steps and Business Questions with Visualization

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Program: Data Analytics with Tableau

Organization: SmartBridge – SkillWallet

Year: 2026

Dataset Overview

The Heart Disease dataset contains patient health information collected from multiple demographic, lifestyle, and medical attributes. The dataset is used to analyse factors contributing to heart disease and identify high-risk patient groups through interactive visualizations.

Dataset Details:

Attribute	Description
Dataset Name	Heart Disease Dataset
File Format	CSV
Total Records	4500
Tool Used	Tableau Public
Domain	Healthcare

Data Preprocessing Steps

The following preprocessing steps were performed before creating the Tableau dashboard:

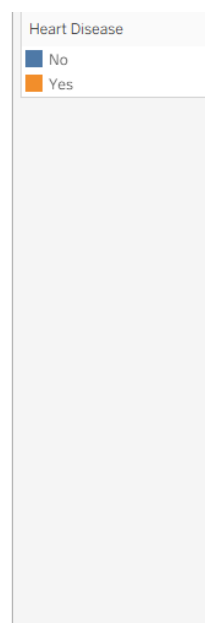
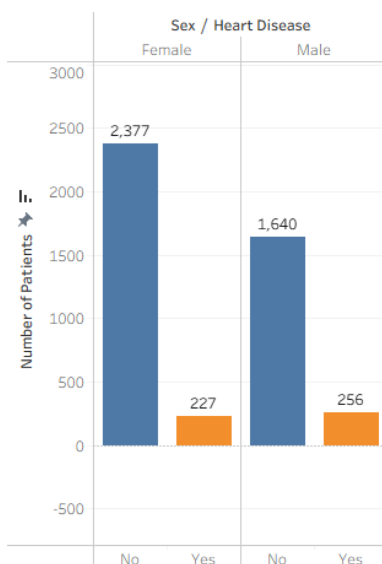
- Imported the CSV dataset into Tableau.
- Verified data types for all attributes.
- Removed duplicate records.
- Checked for missing values.
- Renamed fields wherever required.
- Created calculated fields.
- Organized dimensions and measures.
- Verified consistency of categorical values.
- Prepared data for visualization.

Business Question 1

Question: How does heart disease vary by gender?

Visualization -

Gender-wise Distribution of Heart Disease



Observation:

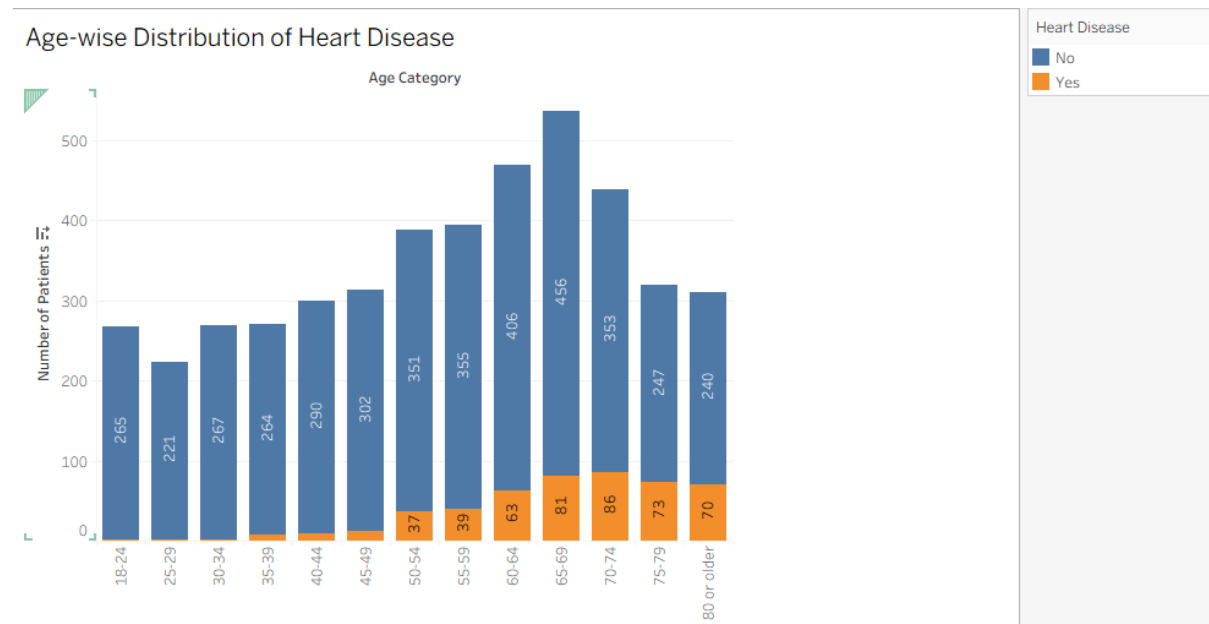
- Male patients show a higher number of heart disease cases than female patients.

- Heart disease is more prevalent among males in the analyzed dataset.
- Gender is an important factor associated with heart disease risk.

Business Question 2

Question: Which age groups are most affected by heart disease?

Visualization –



Observation -

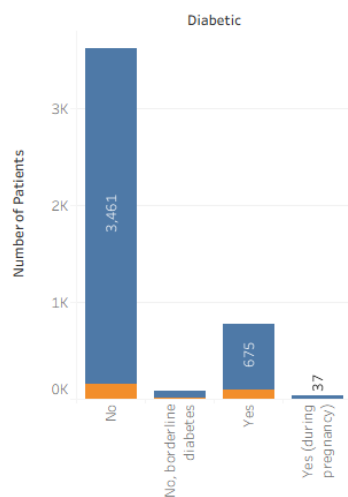
- Heart disease cases increase with age.
- Individuals aged **60 years and above** have the highest number of heart disease cases.
- Older adults are at greater risk and require regular health monitoring.

Business Question 3

Question: What is the relationship between diabetes and stroke?

Visualization –

Stroke Distribution Across Diabetes Status



Observation -

- Stroke cases are more common among diabetic patients than non-diabetic patients.
- Diabetes is an important risk factor that increases the likelihood of stroke.
- Proper diabetes management may help reduce stroke risk.

Business Question 4

Question: How do smoking and alcohol consumption affect heart disease?

Visualization –

Impact of Smoking and Alcohol on Heart Disease

Alcohol Dri..	Smoking	
	No	Yes
No	189	280
Yes	5	9

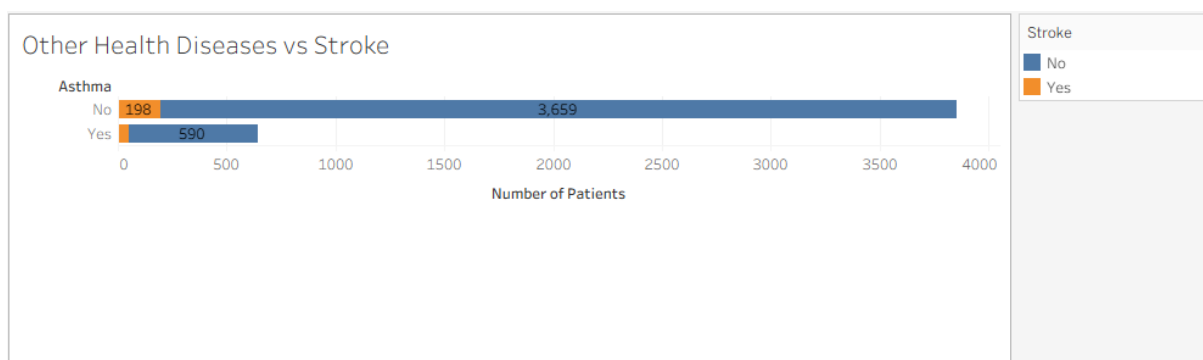
Observation -

- Smokers have a higher occurrence of heart disease than non-smokers.
- Alcohol consumption also contributes to an increased risk of heart disease.
- Adopting healthier lifestyle habits can reduce cardiovascular risk.

Business Question 5

Question: How do other health conditions influence stroke?

Visualization -



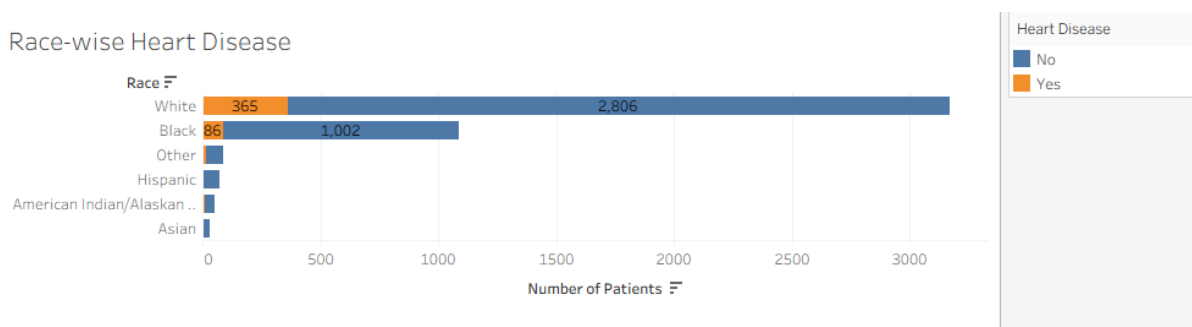
Observation / Answer

- Patients with other health conditions such as asthma or kidney disease are more likely to experience stroke.
- Multiple medical conditions increase overall health risks.
- Early diagnosis and regular monitoring can improve patient outcomes.

Business Question 6

Question: Which race has the highest number of heart disease cases?

Visualization -



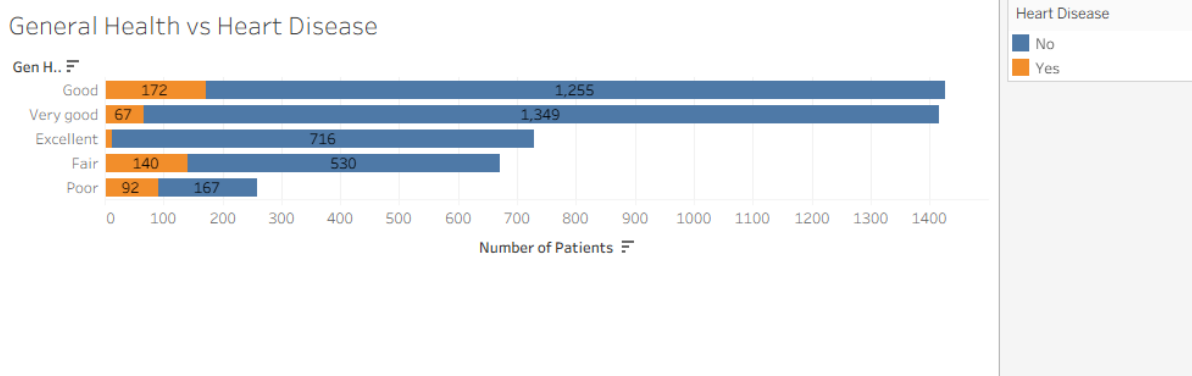
Observation -

- Heart disease prevalence varies across different racial groups.
- Some racial groups have a higher number of reported heart disease cases than others.
- This analysis helps identify population groups that may benefit from targeted healthcare initiatives.

Business Question 7

Question: How does general health affect heart disease?

Visualization -



Observation -

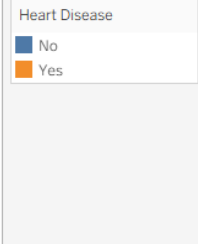
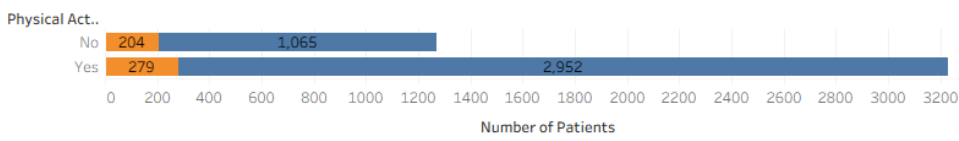
- Individuals with poor general health have a higher number of heart disease cases.
- Better overall health is associated with a lower occurrence of heart disease.
- Maintaining good health can contribute to reducing cardiovascular risk.

Business Question 8

Question: Does physical activity reduce the risk of heart disease?

Visualization -

Physical Activity vs Heart Disease



Observation -

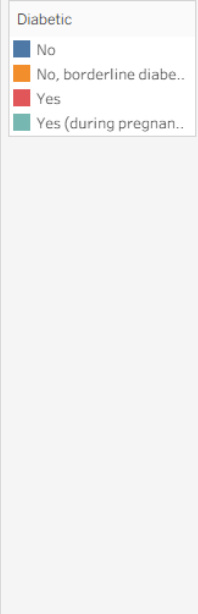
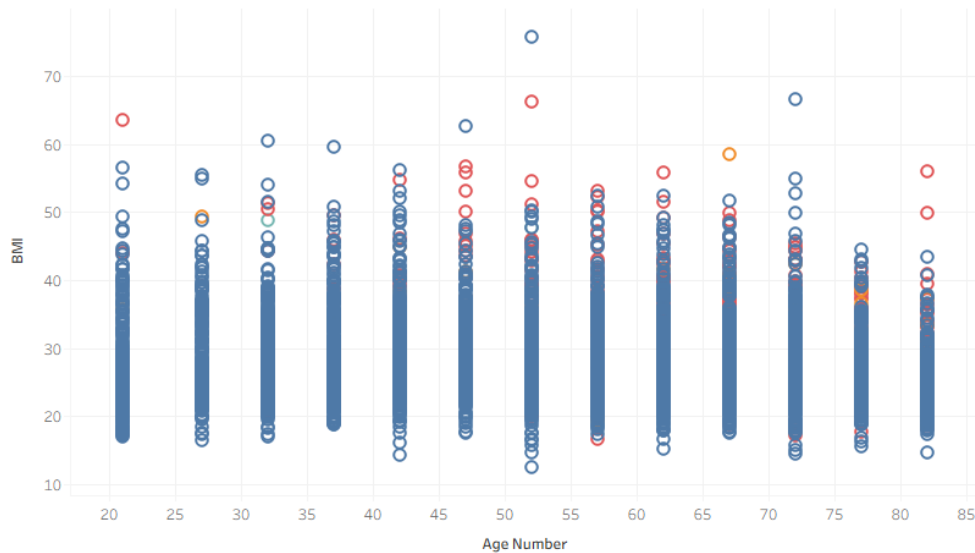
- Individuals who engage in regular physical activity have fewer heart disease cases.
- Lack of physical activity is associated with a higher risk of heart disease.
- Regular exercise plays an important role in maintaining cardiovascular health.

Business Question 9

Question: What is the relationship between age, BMI, and diabetes?

Visualization –

Age and BMI vs Diabetic



Observation -

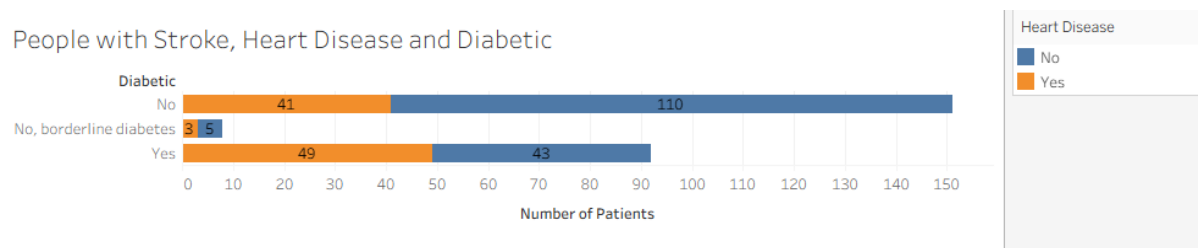
- Higher BMI values are more frequently observed among diabetic patients.
- Increasing age combined with higher BMI is associated with a greater likelihood of diabetes.

- Maintaining a healthy body weight can help lower diabetes-related health risks.

Business Question 10

Question: How many stroke patients also suffer from heart disease and diabetes?

Visualization -



Observation -

- A significant number of stroke patients also have heart disease and diabetes.
- The coexistence of these conditions indicates a high-risk patient group.
- Integrated healthcare management is essential for improving outcomes in patients with multiple chronic conditions.